

Finger Kinematic Modeling and Real-Time Hand Motion Estimation

P. CERVERI,¹ E. DE MOMI,¹ N. LOPOMO,¹ G. BAUD-BOVY,² R. M. L. BARROS,³ and G. FERRIGNO¹

¹Bioengineering Department, Politecnico di Milano University, Piazza Leonardo da Vinci, 32, I-20133 Milan, Italy; ²Faculty of Psychology, San Raffaele Vita-Salute University, and IIT Network Research Unit of Molecular Neuroscience, San Raffaele Foundation, via Olgettina, 58, I-20132 Milan, Italy; and ³Laboratório de Instrumentação para Biomecânica, Faculdade de Educação Física – Universidade Estadual de Campinas, Campinas, Brazil

(Received 28 September 2005; accepted 2 August 2007; published online 15 August 2007)

Abstract—This paper describes methods and experimental studies concerned with quantitative reconstruction of finger movements in real-time, by means of multi-camera system and 24 surface markers. The approach utilizes a kinematic model of the articulated hand which consists in a hierarchical chain of rigid body segments characterized by 22 functional degrees of freedom and the global roto-translation. This work is focused on the experimental evaluation of a kinematical hand model for biomechanical analysis purposes.

From a static posture, a completely automatic calibration procedure, based on anthropometric measures and geometric constraints, computes axes, and centers of rotations which are then utilized as the base of an interactive real-time animation of the hand model. The motion tracking, based on automatic marker labeling and predictive filter, is empowered by introducing constraints from functional finger postures. The validation is performed on four normal subjects through different right-handed motor tasks involving voluntary flexion-extension of the thumb, voluntary abduction-adduction of the thumb, grasping, and finger pointing. Performances are tested in terms of repeatability of angular profiles, model-based ability to predict marker trajectories and tracking success during real-time motion estimation. Results show intra-subject repeatability of the model calibration both to different postures and to re-marking in the range of 0.5 and 2 mm, respectively. Kinematic estimation proves satisfactory in terms of prediction capability (index finger: maximum RMSE 2.02 mm; thumb: maximum RMSE 3.25 mm) and motion reproducibility (R^2 coefficients—index finger: 0.96, thumb: 0.94). During fast grasping sequence (60 Hz), the percentage of residual marker occlusions is less than 1% and processing and visualization frequency of 50 Hz confirms the real-time capability of the motion estimation system.

Keywords—Hand kinematic model, Real-time tracking, Thumb motion, Reaching, grasping, Finger movements.

INTRODUCTION

Since last years the analysis of *in-vivo* hand motion, specifically focused on finger joints, has gained growing attention in several scientific domains like biomechanics,^{3,5,9,14,18,27,34,37} neurophysiology,^{17,20,25,28,32} and hand surgery.^{6,8,33,36} Quantitative evaluation of traumatic and pathologic effects on hand functions, planning of surgical intervention, effectiveness of rehabilitative therapy and physiological understanding represent biomedical applications that can benefit from accurate kinematic and dynamic models of the hand structure. In particular, the ability to replicate in real-time articulated hand movements is advocated by research domains like psychological investigation of multi-sensorial integration, tele-manipulation, device remote control and human-computer interaction.^{22–24,31} However, because of the great number of involved degrees of freedom (DoF) and of the complexity of the mapping between external measurements and internal functional variables, the accurate assessment of *in-vivo* hand kinematics and dynamics represents a complex task.

Electromyography (EMG) and kinetic data are rather common measures for investigating muscle coordination or synergy, but they do not supply accurate information about segmental motion evolution.^{11,28,32} Electromagnetic and video-based systems have been largely proposed in the field of motion analysis and they proved effectiveness for gait analysis, but their use for hand motion analysis has been poorly investigated. While electromagnetic motion analysis systems were applied to measure gross hand movements,¹⁴ the success of video-based systems coupled with surface markers to estimate fine finger motion was demonstrated through fluoroscopy validation, where waveforms obtained from surface markers compared favorably with “gold standard” fluoroscopy waveforms.²⁶ However, only planar movements of the forefinger were analyzed.

Address correspondence to P. Cerveri, Bioengineering Department, Politecnico di Milano University, Piazza Leonardo da Vinci, 32, I-20133 Milan, Italy. Electronic mail: pietro.cerveri@polimi.it

Using two markers placed at the extremities of a little bar, attached to the thumb nail,⁹ presented a method for quantifying the thumb rotation, during circumduction movements, surveyed by a video camera. As a result, they found high correlation between the motor patterns of the left and the right thumb which candidates the technique to assessment of functionality after surgical recovering, using the contro-lateral side as a control. However, the intra-subject variability of the technique was not investigated and the oversimplified kinematic description of the thumb structure through a single axis limited its applicability. Hand motion studies employing video measurement systems have been focused so far to analyses of single finger postures or motions,^{7,8,18,19} fingertip traces, joint flexion-extension initiation and sequence,³⁰ fingers coordination patterns and kinematics synergies.³

Four main problems still remain critical in this research field: kinematic description (joint modeling and number of DoFs), model calibration (which means locating the axes of rotation (AoR) and centers of rotation (CoR) from external measurements, starting from the kinematic description of the model), optimization of marker protocol (marker visibility, correct constraints for motion reconstruction of the defined kinematic description, re-marking), and mapping between the marker motion and the evolution of the joint motion.

The key role of the kinematic description was demonstrated by Cuevas's group showing that, even in presence of accurate assessment of the musculo-skeletal parameters, incorrect kinematic model can dramatically disrupt the results of the biomechanical analysis. They proposed a 5-axis, 8-muscle biomechanical model—the trapezium-metacarpal (TM) joint involving two orthogonal and intersecting axes of rotation—to compare the predicted ranges of maximal static force production and muscle activity to experimental measurements of maximal voluntary thumb-tip force and EMG.³² Obtained predictions were unrealistic with raw coordination patterns. While the sensitivity of the model to biomechanical parameters proved to be less critical, inadequate kinematic description was highlighted to be the main reason of failure.

Different authors proposed methods for the computation of CoR locations and AoR directions based on surface marker measurements.^{5,10,37} In particular,³⁷ developed an analytical method to compute the CoRs from twenty-one markers for 1 DoF joints during flexion-extension movements. Effects of small range of motion (RoM), which were shown to significantly affect the estimated kinematic variables, were not explored.²¹ In Ref. 5, the authors proposed a method based on forty-two markers to estimate the AoR

directions and the CoR locations of the whole hand complex. While the technique was robust to small motion ranges, the protocol was too cumbersome, the trapezium-metacarpal (TM) joint kinematics was approximated by a spherical joint and the real-time motion assessment was not carried out.

This work proposes a technical approach attempting at overcoming some of the previous limitations. The main objectives are the experimental evaluation of a kinematical hand model for biomechanical analysis purposes and the real-time accurate hand motion estimation. The proposed kinematical model consists in a hierarchical chain of rigid body segments characterized by twenty-two functional degrees of freedom (with particular attention to the TM joint), whose motion evolution is computed using twenty-four markers. A straightforward calibration procedure, based on anthropometric measures and geometric constraints, computes the AoR and the CoR from a reference hand posture. The motion tracking, resting on automatic marker labeling and on predictive filter, is empowered by introducing constraints from functional finger postures. All such functionalities were integrated within application software, which provides real-time hand animation in a 3D graphic environment. Validation, in terms of calibration, motion assessment repeatability and real-time performance, is carried out on four normal subjects across different articulated right-handed movements.

METHODOLOGICAL FRAMEWORK

Hand Skeleton Model and Marker Protocol

In order to represent the hand skeleton into a kinematic chain we use a hierarchical structure of rigidly connected joints (Fig. 1a), each having one (pivot joint) or two (cardan joint) rotational DoFs. A consistent approximation consists on representing the 8 bones of the carpus, which finely articulate with each other, into a single rigid segment.^{1,4,35} Grouping the four metacarpal bones in the hand dorsum into a single segment represents a good compromise between complexity and accuracy of the representation.^{5,37} Two functional DoFs, namely flexion-extension and abduction-adduction, characterize the wrist joint. One functional DoF for flexion-extension characterizes both proximal (PIP) and distal intra-phalangeal (DIP) joints. Two functional DoFs account for the kinematics of the metacarpo-phalangeal (MCP) joints. The thumb has three joints: trapeziometacarpal (TM), metacarpo-phalangeal (MCP), and interphalangeal (IP). TM and MCP have 2 DoFs and IP 1 DoF. The kinematic description of the TM joint needs additional

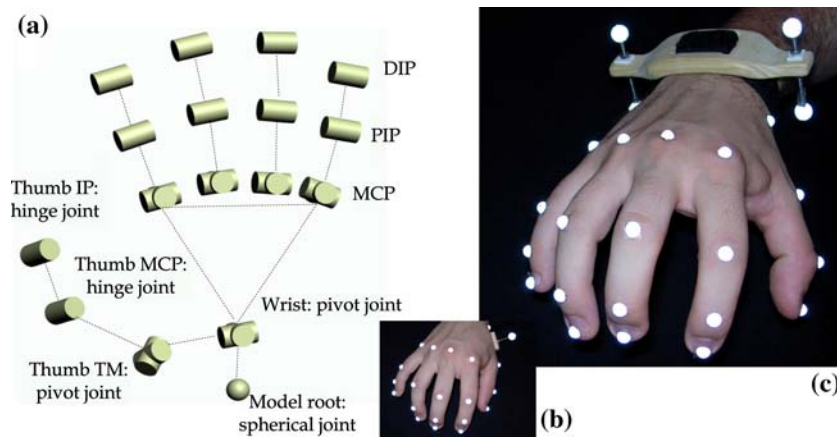


FIGURE 1. Kinematic model of the complete hand structure: the model root is characterized by a spherical joint endowed with three rotational DoFs and three translations. The wrist is modeled with a pivot joint (two orthogonal rotation axes—flexion-extension and adduction-abduction). The TM joint is represented by two coupled hinge joints; IP and DIP joints of the thumb are modeled with hinge joints (flexion-extension); the MCP of the fingers is modeled with a pivot joint (flexion-extension and adduction-abduction); the PIP and DIP joints of the fingers are modeled with hinge joints (flexion-extension) (a). Additional marker used to calibrate thumb intrinsic pronation (c): it is positioned on a rigid structure bound to the first metacarpal segment and directed outward from a flat open palm. With the two markers on TM and MCP, it defines a fit plane whose normal characterizes the flexion X-axis; the adduction Z-axis is orthogonal to X-axis, belongs to the plane and passes through the TM marker. The pronation axis is therefore estimated as $Z \times X$ cross product (b). Configuration of 24 markers used in the marker protocol: four markers, rigidly linked on a structure locked on the wrist, account for the global translation and orientation of the hand. Each finger is endowed with four markers placed in correspondence of joints. The four markers in correspondence of the metacarpo-phalangeal joints allow the palm motion reconstruction (c).

attention: it has in fact three resulting types of motion (flexion-extension, abduction-adduction and pronation-supination), but it only has two DoFs. In other words, for a given flexion and abduction, the thumb has a unique pronation (twist). It was demonstrated that this is because the thumb physiological rotation axes are non-orthogonal and non-intersecting.^{12,13,18,29} In the model calibration stage, we take into account thumb static pronation angle with an additional marker in order to compute the physiological direction of the adduction axis (Fig. 1b).

To compute the parameters of the proposed model (model calibration), i.e., the relative position of the joint centers and the directions of the rotation axes, a configuration of twenty-four markers is designed (Fig. 1c). In order to compute the wrist coordinate frame and the global position and orientation of the model, we adopted a noose with 4 markers rigidly linked with each other (two approximately aligned to the ulna styloid and to the radius styloid processes and the other two placed at the extremes of two small sticks fixed to the noose). In order to compute the hand dorsum coordinate frame and kinematics, four markers are placed on the hand torso in correspondence of MCPs. Three markers are used for each finger: one marker for PIP and DIP joints each and one marker on the finger tip on the distal border the nail (FT). The thumb is fit with 4 markers: one is placed at the base of the metacarpal segment, between the tendons of the short and long adductors of the metacarpal where the

effects of skin stretching are minimal, two are placed at the proximal and at the distal phalanx, respectively, and one is placed at the fingertip. In order to calibrate thumb intrinsic pronation, an additional marker is used: it is bound to a rigid structure placed at the middle of the metacarpal segment not being aligned with the markers on the TM and MCP joints and elevating towards hand dorsum along an ideal adduction axis.

Marker Labeling and Model Calibration

In order to scale the model parameters to the real hand of the subject, a controlled static posture, with the palm aligned with the wrist and the fingers slightly bent (Fig. 1c), is considered. Markers labeling is started by searching the pairs of markers whose inter-distances are equal to the vertical and the horizontal distances between the markers rigidly linked at the wrist bracelet (Fig. 2a) and then by checking co-planarity among the four markers. The geometric center (GC) of such makers is then computed. The algorithm then labels thumb markers (Fig. 2b) with an iterative search for the marker closer to the wrist GC. Then MCPs are recognized as the four markers closer to the wrist GC and labeled by increasing distances from thumb MCP (Fig. 2c); remaining markers on the fingers are projected along the line joining the first and the fourth MCP markers (the origin is set as pinky MCP) and grouped three by three according to scalar

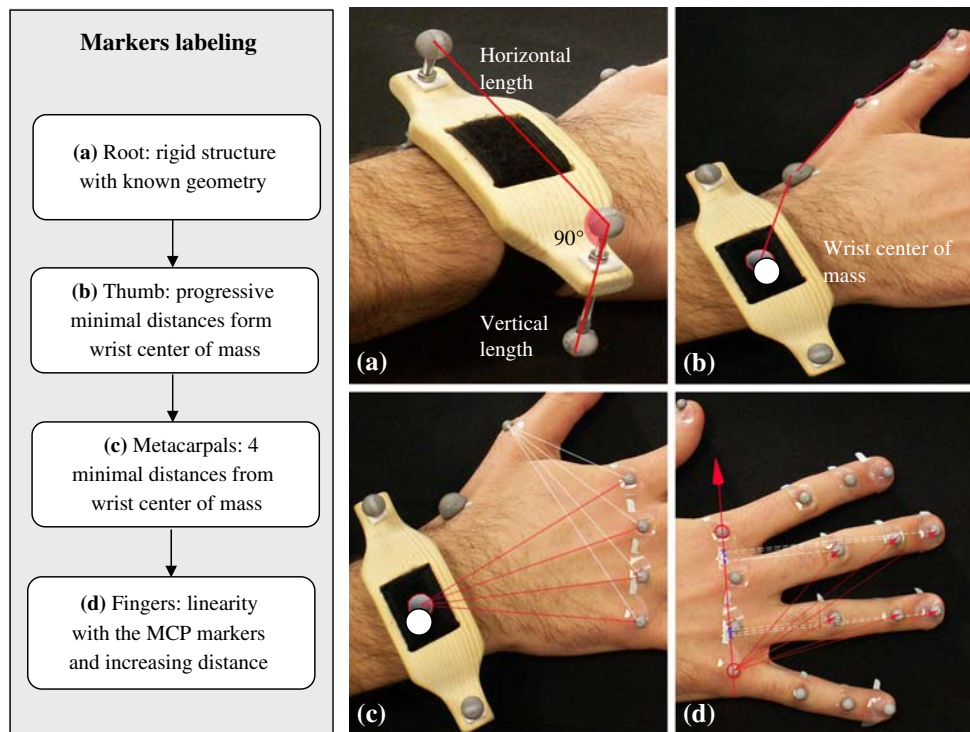


FIGURE 2. Chart of the automatic labeling procedure. The markers on the wrist are first classified using geometrical constraints originated from the structure geometry (a). Then thumb markers are recovered using progressive distance constraints from wrist markers (b). Markers at the metacarpo-phalangeal joints are classified using minimal distances from wrist markers (c). At last, each finger is labeled using geometric constraints (d).

product values. The markers belonging to one finger are finally labeled using the increasing distance from the corresponding marker at MCP (Fig. 2d).

After marker labeling, AoR and CoR are computed based on four anthropometric measures specific for each subject hand, namely the TM, MCP, PIP and DIP joint thickness, as explained below. The root absolute coordinate system is positioned in the GC of the 4 markers at the wrist with the *Y*-axis defined as the 4 markers plane normal, the *Z*-axis as the direction connecting two markers vertically and rigidly linked with each other and the *X*-axis with the right hand rule. The wrist coordinate system is obtained defining the *Y*-axis as the direction connecting the wrist center with the marker placed on the middle finger MCP joint, the *Z*-axis as the palm plane normal direction, the *X*-axis with the right hand rule. Apart from the thumb, each proximal phalanx coordinate system has the *X*-axis directed as the normal direction to the plane defined by the 4 markers on MCP, PIP, DIP, and FT of the corresponding finger. Medial and distal phalanges coordinate systems are obtained by inheriting the *X*-axis from their corresponding MCP district. The *X*-axis of the TM coordinate system is directed as the normal direction to the plane defined by the 4 markers

at TM, MCP, IP and FT. The estimation of the *Z*-axis orientation of TM coordinate system, i.e., the intrinsic pronation respect to the palm, is obtained thanks to an additional marker, M_a , positioned on a rigid structure bound to the metacarpal segment and directed outward from a flat open palm (Fig. 1b).

The origins of the coordinate systems of the finger phalanges are positioned in points obtained by translating the corresponding marker coordinates of half the joint thickness along the local *Z*-axis direction. An example of calibrated kinematic model is displayed in Fig. 3.

Marker Tracking and Motion Estimation

During hand motion, forward kinematics approach is pursued by mapping the 3D measured coordinates, expressed in the absolute coordinate system, to the corresponding coordinates computed in the calibration stage, expressed in the local coordinate systems. Frame by frame, a dynamic marker labeling procedure utilizes a short-time memory of the marker trajectories to predict marker movements, thus increasing the labeling reliability. The prediction is utilized to dynamically bind reconstructed markers to the corresponding

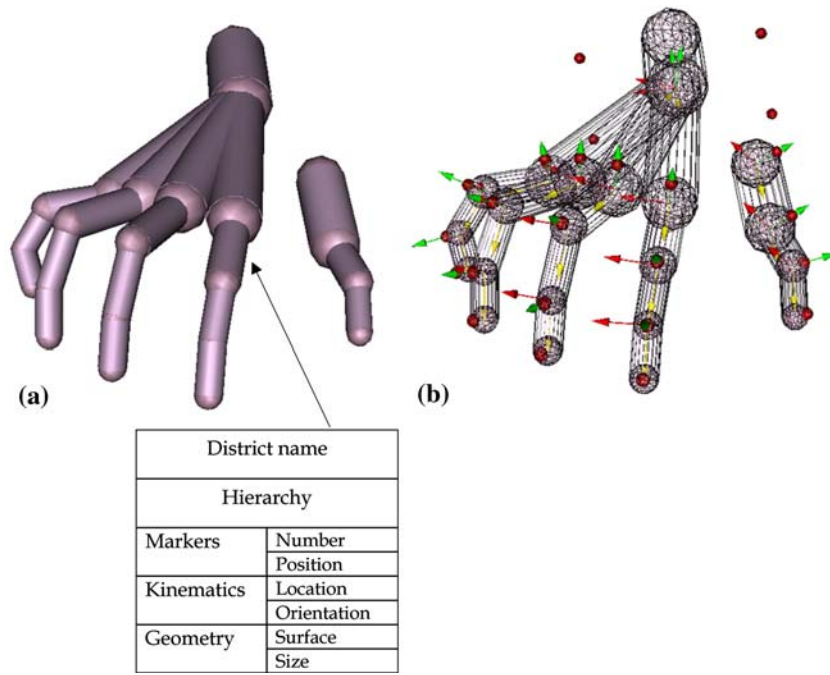


FIGURE 3. Hand model presented as surface (a) and wire frame (b) view. The model consists of spheres which reproduce joints and cylinders which reproduce bone segments. Each anatomical district is characterized by name, its position in the kinematical chain, which represents hierarchy information, markers position and number, its kinematical variables (i.e., active DoFs, absolute and local rototraslation matrices and anthropometric measures) and its geometrical shapes (number, type, position, orientation). The calibration process includes the computation of geometric and kinematic characteristics, done using anthropometric measures taken on the subject and the marker positions reconstructed and labeled in the calibration posture.

trajectory, using adaptive proximity thresholds. When the success of the dynamic marker labeling is compromised, the algorithm deletes the history of the marker trajectory and automatically switches to the static labeling procedure (Fig. 4).

Since articulated hand postures may produce markers occlusion to cameras, criteria based on geometrical and functional constraints are integrated to recover the position of occluded markers and to increase the robustness of the motion estimation (Fig. 5). Knowing the inter-distances (k_u , k_v) among the 4 markers at the wrist allows tolerating one of those markers occlusion (Fig. 5a). Positions of the markers at the MCP finger joint, which defines the palm kinematics, are related by distances (r_i) and angular displacements (α , β , γ) in the corresponding local coordinate system (Fig. 5b). The 3D coordinates of any two occluded markers can be reconstructed starting from the coordinates of two other markers. Four markers provide the phalanx kinematics: three of them and relations of angular joints are sufficient to locate the missing fourth (Figs. 5c, 5d).

The expression of motion evolution requires the computation of the global pose of the root (as in the calibration stage) and the update of the angular pose of each anatomical district from the previous posture (Fig. 6). Since each anatomical district of the model is

individuated by one or two markers,¹ the computation of one or two DoFs motion requires information about the kinematic chain. Assuming that the local coordinates of the markers do not vary during motion (rigid-body assumption), the 1 DoF motion (e.g., flexion-extension of the middle phalanx) at time instant t is solved from the following relation:

$$\mathbf{P}_p = \mathbf{R}_{1\text{DoF}}(X, \alpha) \mathbf{P}_l \quad (1)$$

where $\mathbf{P}_p$ and $\mathbf{P}_l$ are the reconstructed marker coordinates, expressed in the proximal and local districts of the model, respectively, and $\mathbf{R}_{1\text{DoF}}(X, \alpha)$ is the rotation matrix accounting for rotation α around the local X -axis. By developing the matrix product in Eq. (1) as:

$$\begin{pmatrix} x_p \\ y_p \\ z_p \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(\alpha) & -\sin(\alpha) \\ 0 & \sin(\alpha) & \cos(\alpha) \end{pmatrix} \begin{pmatrix} x_l \\ y_l \\ z_l \end{pmatrix} = \begin{pmatrix} x_l \\ y_l \cos(\alpha) - z_l \sin(\alpha) \\ y_l \sin(\alpha) + z_l \cos(\alpha) \end{pmatrix} \quad (2)$$

¹ The palm rotation (2 DoFs) is determined by the two markers at MCP external surface of the index and middle fingers. Apart from the thumb, the rotation of the MCP (2 DoFs), PIP (1 DoF) and DIP (1 DoF) joints of the fingers are all computed by a single marker.

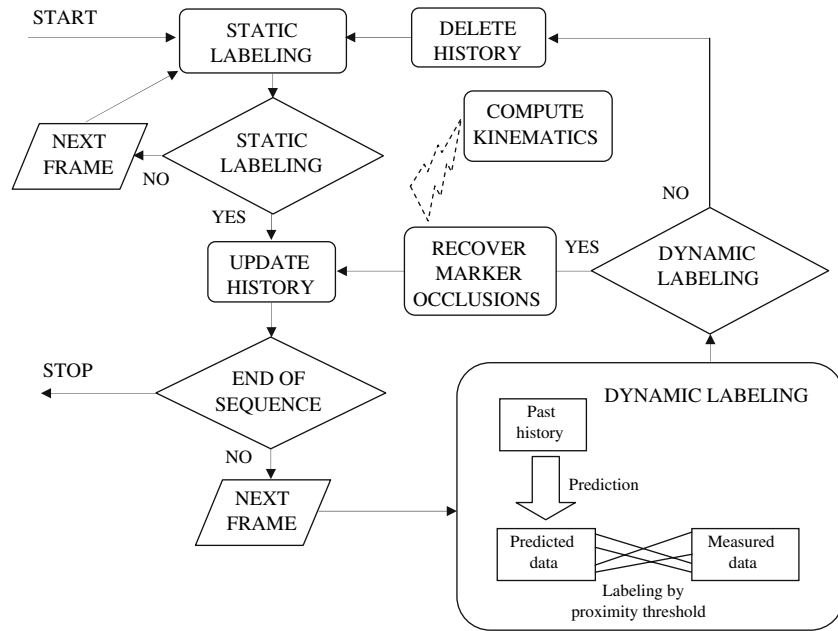


FIGURE 4. Flow chart of the marker tracking procedure. It consists on interaction between static and dynamic labeling algorithms. When the marker trajectory history is built (two consecutive tracked frames are sufficient) then data processing is switched to dynamic labeling. Marker trajectory prediction is utilized to label current measured markers. The labeling is performed by two following proximity thresholds which represent the search radius around the prediction. Then, geometric and functional constraints are utilized to recover potentially occluded markers. If dynamic labeling fails the processing flow deletes the frame history and switches to static labeling.

the α angle is computed by solving the 2^o-order algebraic equation.

The 2 DoFs rotation $\mathbf{R}_{2\text{DoFs}}(X, \alpha; Z, \beta)$ around two intersecting and orthogonal axes (e.g., flexion-extension and adduction-abduction of the proximal phalanx) is solved by first considering the flexion-extension, $\mathbf{R}_{1\text{DoF}}(X, \alpha)$, and then the adduction-abduction, $\mathbf{R}_{1\text{DoF}}(Z, \beta)$. The following relation is utilized to estimate the two rotation angles α and β :

$$\begin{aligned} \mathbf{P}_p &= \mathbf{R}_{2\text{DoFs}}(X, \alpha; Z, \beta) \\ \cdot \mathbf{P}_l &= \mathbf{R}_{1\text{DoF}}(Z, \beta) \mathbf{R}_{1\text{DoF}}(X, \alpha) \cdot \mathbf{P}_l \end{aligned} \quad (3)$$

where $\mathbf{P}_p$ and $\mathbf{P}_l$ are the reconstructed marker coordinates, expressed in the proximal and the local districts of the model, respectively. Equation (3) is expanded as:

$$\begin{aligned} X_p &= \cos(\beta)X_l - \sin(\beta)\cos(\alpha)Y_l + \sin(\beta)\sin(\alpha)Z_l \\ Y_p &= \sin(\beta)X_l + \cos(\beta)\cos(\alpha)Y_l - \cos(\beta)\sin(\alpha)Z_l \\ Z_p &= \sin(\alpha)Y_l + \cos(\alpha)Z_l \end{aligned} \quad (4)$$

After computing α angle from the third equation and substituting it in the second equation, β angle is obtained. In order to avoid unnatural hand motion, we implement heuristic relations among joint angles, such as: $0^\circ \leq \theta_{\text{MCP-FE}} \leq 90^\circ$, $-15^\circ \leq \theta_{\text{MCP-AA}} \leq 15^\circ$, $0^\circ \leq \theta_{\text{PIP}} \leq 110^\circ$, $0^\circ \leq \theta_{\text{DIP}} \leq 90^\circ$. The motion of DIP and

PIP joints is not independent and can be described as $\theta_{\text{DIP}} = 2/3 \theta_{\text{PIP}}$.¹⁵ We finally smooth angular values to reduce the 3D reconstruction noise through FIR real-time low pass filters tuned with the acquisition frequency.

EXPERIMENTAL FRAMEWORK

Data were acquired by six cameras (SMART by BTS, Milan—Italy). Camera calibration accuracy (0.30 mm) was verified by measuring the RMS error of the distribution of the reconstructed distance between two rigidly linked markers, at known distance, acquired in several positions within the working volume (70 cm × 50 cm × 70 cm).

Experimental Session—Model Calibration Repeatability

Tests

Intra-subject repeatability of both model calibration and re-marking procedures were considered. In the first trial, each subject was asked to place his/her hand in the working volume in the calibration posture aligning the palm with the wrist and keeping the fingers

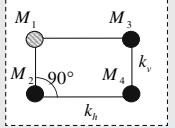
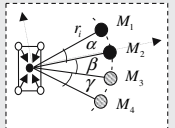
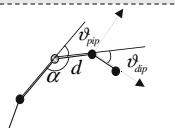
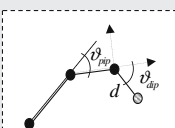
DISTRICT	GEOMETRICAL AND FUNCTIONAL RELATIONS
Wrist	 $M_1 = M_2 + k_v \frac{(M_3 - M_4)}{k_h}$ $M_4 = M_3 + k_h \frac{(M_2 - M_1)}{k_v}$ (a)
Palm	 $M_1^{Loc}(\rho, \vartheta) = (r_1, \alpha)$ $M_2^{Loc}(\rho, \vartheta) = (r_2, 0)$ $M_3^{Loc}(\rho, \vartheta) = (r_3, -\beta)$ $M_4^{Loc}(\rho, \vartheta) = (r_4, -\beta - \gamma)$ (b)
Proximal phalanges	 $\vartheta_{pip} = 180^\circ - \alpha$ $\vartheta_{dip} = \frac{2}{3} \vartheta_{pip}$ $M_{pip}^{Loc}(\rho, \vartheta) = (d, 180 + \vartheta_{dip})$ (c)
Distal phalanges	 $\vartheta_{dip} = \frac{2}{3} \vartheta_{pip}$ $M_{ip}^{Loc}(\rho, \vartheta) = (d, \vartheta_{dip})$ (d)

FIGURE 5. Criteria to increase the labeling robustness to marker occlusion and the hand motion tracking procedures. One occluded marker among the four ones at the wrist can be virtually reconstructed from the other three ones. Two occluded markers at MCP joints can be reconstructed using geometric constraints computed in the calibration stage. For the fingers, any one occluded marker can be reconstructed from the other two applying geometric and functional constraints on the joint angular ranges.

slightly bent. Small finger displacements were allowed with the palm position fixed. Fingertips range of motion was approximately 1 cm in translation. 25 different repetitions of the same reference posture were acquired and processed. In the second trial, each subject was marked 25 different times by an expert operator and data were acquired in each single calibration posture.

Performance Indices

The standard deviation of the segmental length, computed as the distance between two consecutive CoRs,² $^i d_k$ (mm) of the k -th anatomical district of the i -th subject was computed over $N = 25$ calibration sessions. Without lack of generality, we focused the analysis on the following segmental lengths representative for all digits: first metacarpus, thumb proximal and distal phalanges, index proximal, medial and distal phalanges. The standard deviation $^i a$ (degrees) of the AoR directions, in the corresponding local coordinate

system, was computed over $N = 25$ calibration sessions.

Experimental Session—Analysis of the Thumb-Index Finger Motion

Tests

Subjects performed four types of movement tasks during this session: (1) voluntary flexion-extension of the thumb, (2) voluntary abduction-adduction of the thumb, (3) tip-to-tip grasping and (4) cylinder. All the movements started with the forearm horizontal and the fingers in natural full extension with the palm facing medially. In the first task, subjects were asked to lift the hand from the base at approximately 20 cm, and to perform 15 repetitions of adduction-abduction of the thumb starting from the maximum adduction angle to a position where the thumb got in touch with the index, without consciously controlling involuntary pronation of the metacarpal and avoiding voluntary flexion of the phalanges.

In the second task, subjects were asked to keep the same initial posture as in the first task (maximum adduction and extension without extra-extension) and to perform 15 consecutive flexion-extensions of the thumb metacarpal and phalanges. Subjects were asked to flex the metacarpal about the palm of about 90°.

In the third task, subjects performed 15 consecutive repetitions of tip-to-tip grasping with the dorsum turned upright, involving flexion-extension and adduction-abduction of both the index and the thumb.

For the latter task, subjects reached forward over a distance of approximately 20 cm to grasp a vertical cylindrical handle (7.5 cm in diameter) by concurrent voluntary flexion of all digits, returned to the initial posture, paused and repeated the grasping 15 times. It has to be noted that arm transport motion was performed without significant forearm pronation-supination and torso assistance.

We derived and processed separately angular profiles for index MCP, PIP, DIP and thumb TM, MCP, IP joints. Resting time was allowed to the subjects. Each repetition had to be synchronized with a 2-second beat time, in order to allow time normalization of the profiles. A cubic spline interpolant, whose knots are defined by the relative minimums (the initial and final values) of each repetition (lasting approximately 1 sec), was computed for each profile baseline. It was then subtracted to the profile to compensate for baseline shift over time, reducing the baseline oscillations and allowing angular ranges comparison.

In order to predict trajectories by the model, kinematic was computed between two consecutive markers positions thanks to Eqs. (1)–(4).

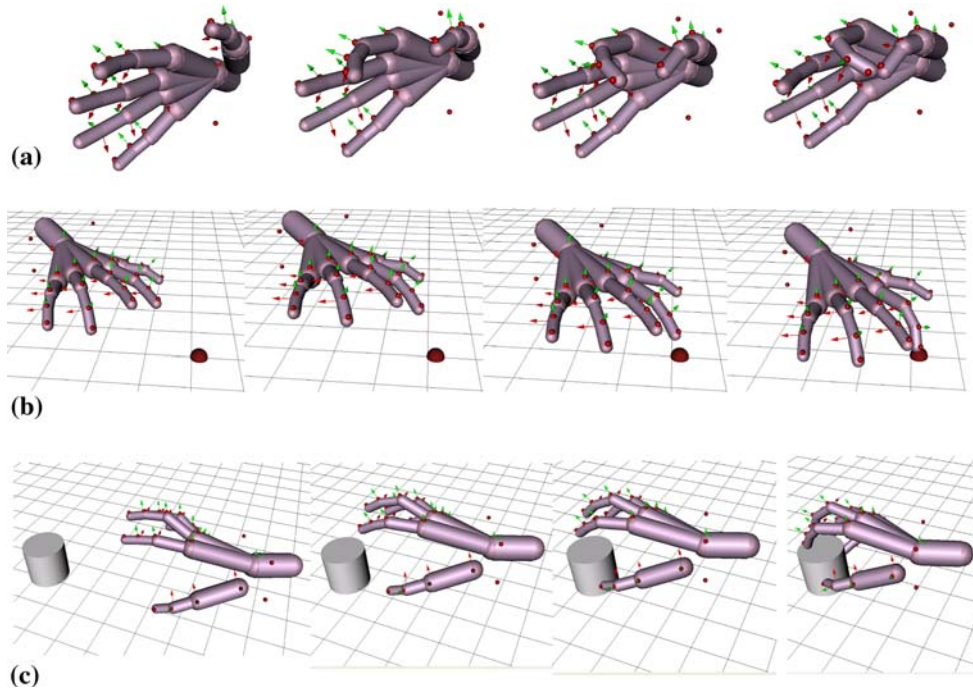


FIGURE 6. Snapshots of three hand motion sequences: tip-to-tip grasping (a), index finger pointing (b), cylinder grasping (c).

Performance Indices

In order to quantify the accuracy of the kinematics description, we first computed Δ_r (for each movement repetition r) as:

$$\Delta_r = \sqrt{\frac{\sum_i (\bar{P}_i - P_i)^2}{N}} \quad (5)$$

where $\bar{P}_i$ and P_i are the marker point predicted by the model and that measured by the motion capture system, respectively, and N is the number of frames. We reported the accuracy, Δ , computed across M repetitions of the same motion pattern as:

$$\Delta = \sqrt{\frac{\sum_r (\Delta_r)^2}{M}} \quad (6)$$

In order to quantify the repeatability of reconstructed motion pattern,¹⁶ we characterized angular profiles through the Bravais–Pearson coefficient of correlation. We focused this analysis upon tip-to-tip grasping motion pattern. Considering 15 motion repetitions, the profile of each angular variable is associated to a 15 by 15 matrix $\mathbf{C} = \{R_{i,j}^2\}$, with $i, j = 1, \dots, 15$ ($R_{i,j}^2 = R_{j,i}^2$, $R_{i,i}^2 = 1$). The (i, j) entry expresses how the angular profile, estimated from motion repetition i , is linearly correlated with motion repetition j . For each angular profile, we reported the

minimum Bravais–Pearson coefficient R^2 in the matrix $\mathbf{C}$.

Experimental Session—Marker Labeling Performance Tests

Without lack of generality, marker-labeling performance is tested through 4 different motion patterns performed by the first subject. In the first task, the subject was told to maintain his hand on the base and to simulate fast keyboard typing with flexion–extensions of all five digits. In the second task, the subject performed 15 repetitions of tip-to-tip grasping, similar to the one described in the II experimental session, but performed faster, as if it was a reflex movement. In the third task, the subject performed a slow thumb circumduction. In the fourth task, the subject performed 15 fast repetitions of cylinder grasping as described in the II experimental session. All movements were repeated at both 60- and 120-Hz acquisition frequency.

Performance Indices

No marker reconstruction (possible consequence of marker occlusion) may compromise the motion estimation process. Quantification of the labeling algorithm performance is done in terms of percentage of number of lost markers with respect to the number of markers ideally reconstructed along each sequence.

Experimental Session—Real-Time Performance

Tests

The possibility to reproduce acquired movements in real-time within a graphic environment requires to test the computational cost of the implemented algorithms. Equivalently to the previous experimental session, four different motion tasks (fast typing, fast tip-to-tip grasping, slow thumb motion, fast cylinder grasping) were measured. Real-time performances were measured using a Pentium 4 CPU 2.8 GHz, 512-Mb RAM, and NVidia G-Force FX video card.

Performance Indices

Computational cost per frame t is constituted of three parts: firstly, the time t_{3D} needed by the optoelectronic system to reconstruct the 3D coordinates (for the SMART system we measured approximately 1 ms on average for a frame); secondly, the time t_{lm} needed for marker labeling and the motion estimation process; thirdly, the rendering time t_r needed by the graphic engine to refresh all the 3D graphic objects on the virtual scene. In order to quantify the computational cost difference between static or dynamic labeling, analysis of the fast typing task was carried out alternatively using only the static and only the dynamic labeling algorithm.

RESULTS

Model Calibration Repeatability

RMS error values confirm high intra-subject repeatability of the calibration procedure (Table 1,

columns: A). Greater errors are reported for the estimation of the lengths of the thumb phalanges of all subjects with respect to the lengths of the other phalanges of the same subject. For the phalanges of the index finger, the maximum and the minimum RMSE correspond to the medial phalanx of subject 1 and the medial phalanx of subject 4, respectively. Re-marking analysis (Table 1, A, columns: B) shows that the model calibration repeatability decreases by a factor of about 500% with respect to the calibration repeatability across different postures. For AoR direction estimation, without lack of generality, we report results about MCP and IP flexion-extension axes of the thumb and MCP, PIP and DIP flexion-extension axes of the index finger (Table 1B). Both results on segmental length and on AoR directions suggest that that thumb marking is more critical than finger marking.

Analysis of the Thumb-Index Finger Motion

Analysis on the markers of the index finger shows increasing inaccuracy from proximal to distal phalanges in both the reported motor tasks for all the subjects (Table 2). Lower and greater errors in the cylinder-grasping task can be detected at PIP marker and FT marker, respectively. Equivalently, in the tip-to-tip grasping, lower and greater errors can be detected at PIP marker and FT marker, respectively. As expected, we found that the maximum errors occur in correspondence of the maximum flexion RoM. In addition, there is a significant difference between the accuracy computed in the cylinder grasping with respect to the accuracy in the tip-to-tip grasping. The first is higher than the second by a percentage approximately 25% for PIP markers and 20% for DIP

TABLE 1. (A) Model calibration repeatability in terms of standard deviation error (mm) computed on segmental lengths: single marking-different postures (A), remarking-same posture (B).

		Subject 1		Subject 2		Subject 3		Subject 4	
		A	B	A	B	A	B	A	B
(A) Segmental lengths (mm)									
First Metacarpus		0.30	1.48	0.20	1.31	0.15	1.44	0.22	1.50
Thumb Proximal Phalanx		0.28	1.43	0.18	1.36	0.14	1.24	0.27	1.38
Thumb Distal Phalanx		0.27	1.27	0.22	1.18	0.16	0.83	0.06	1.23
Index Proximal Phalanx		0.23	1.33	0.08	0.98	0.15	1.19	0.07	1.23
Index Medial Phalanx		0.27	1.21	0.10	1.08	0.21	1.10	0.03	1.13
Index Distal Phalanx		0.12	1.01	0.12	0.88	0.08	1.12	0.06	1.26
(B) AoR direction estimation (°)									
		Subject 1		Subject 2		Subject 3		Subject 4	
MCP	flexion-extension axes	3.1		3.7		3.4		3.4	
IP		3.4		3.4		2.8		3.0	
MCP	flexion-extension axes	1.9		2.6		2.6		2.3	
PIP		2.0		2.5		2.6		2.5	
DIP		2.0		2.7		2.4		2.4	

(B) Model calibration repeatability in terms of standard deviation error (°) in AoR direction estimation.

TABLE 2. Accuracy of the kinematic description and model calibration in terms of RMS error (mm) of the distance between marker predicted by the model and the one measured by the capture system.

Trial	Subject	Index finger markers (mm)		
		mrk1: PIP	mrk2: DIP	mrk3: FT
15-Repetition cylinder grasping	1	0.42	1.14	1.41
	2	0.65	1.28	1.49
	3	0.75	1.41	1.66
	4	0.68	1.34	1.58
15-Repetition tip-to-tip grasping	1	0.66	1.35	1.72
	2	0.88	1.43	1.83
	3	1.01	1.71	2.02
	4	0.97	1.64	1.97

Presented results refer to the markers of the index finger.

and FT markers. This difference depends on the angular ranges to which joints are subdued: for instance, MCP joint flexion angle ranges about 60° degrees during tip-to-tip grasping and 20° during cylinder grasping. In this case, we can guess that soft tissue artifacts and hand deformation at higher angular ranges, which disrupt the rigid body assumption, can introduce systematic errors.

Markers of the thumb show lower accuracy than the index finger markers (Table 3). Modeling of TM joint is critical and it is witnessed by higher errors in the prediction of the thumb markers. Motor tasks that express great articulation of the TM joint, like tip-to-tip grasping movement, report accuracy decrease with respect to thumb abduction-adduction task. As it can be easily argued, natural motion of the interface of the

first metacarpus with the trapezium is not an ideal pivot joint and the instantaneous rotation center undergone a translation which disrupt our model. Lower errors are reported for MCP markers (thumb abduction-adduction) but the effect of the error propagation on the kinematic chain is clearly noticeable upon DIP (thumb abduction-adduction) and FT (thumb abduction-adduction) markers. Sensible accuracy decrease can be registered when analyzing flexion-extension (FT marker) and tip-to-tip grasping (FT marker) movements.

In Fig. 7, we report flexion-extension angular trajectories of tip-to-tip grasping motion pattern (two cycles). Physiological inter-subject variability is however in agreement with RoM published in literature.¹³ Intra-subject repeatability results are summarized in Tables 4, 5. R^2 coefficient of correlation is 0.96 for all index finger joints, 0.95 for the TM joint and 0.94 for thumb MCP and PIP joints. These values show high reproducibility of the motion patterns.

Marker Labeling Performance

For each task acquired at sampling frequency of 60 or 120 Hz, both the wrist and the palm have the lowest percentage of lost markers, since little movements affect the corresponding markers. The thumb and the index show instead higher percentages of lost marker for fast movements acquired at 60 Hz (task 1: thumb 0.31%, index 11.42%; task 2: 4.02% and 5.87%; task 4: 1.56% and 2.53%) whereas in task 3 (slow thumb motion) percentage of lost markers on the thumb decreases to

TABLE 3. Accuracy of the kinematic description and model calibration in terms of RMS error (mm) of the distance between marker predicted by the model and that one measured by the capture system.

Trial	Subject	Thumb markers (mm)			
		mrk1: TM	mrk2: MCP	mrk3: DIP	mrk4: FT
15-Repetition thumb abduction-adduction	1	1.57	0.47	0.93	1.42
	2	1.70	0.58	1.20	2.12
	3	1.86	0.80	1.80	2.34
	4	1.77	0.74	1.43	1.71
15-Repetition thumb flexion-extension	1	2.12	0.77	2.40	2.66
	2	2.43	1.47	2.35	2.76
	3	2.57	1.65	2.42	2.85
	4	2.44	1.63	2.32	2.80
15-Repetition tip-to-tip grasping	1	2.26	1.05	2.47	2.66
	2	2.53	1.39	2.42	2.91
	3	2.78	1.53	2.78	3.25
	4	2.54	1.48	2.52	3.05
15-Repetition cylinder grasping	1	1.99	0.57	1.19	1.99
	2	2.05	0.66	1.38	2.11
	3	2.12	0.93	1.51	2.17
	4	2.01	0.61	1.26	2.18

Presented results refer to the markers of the thumb.

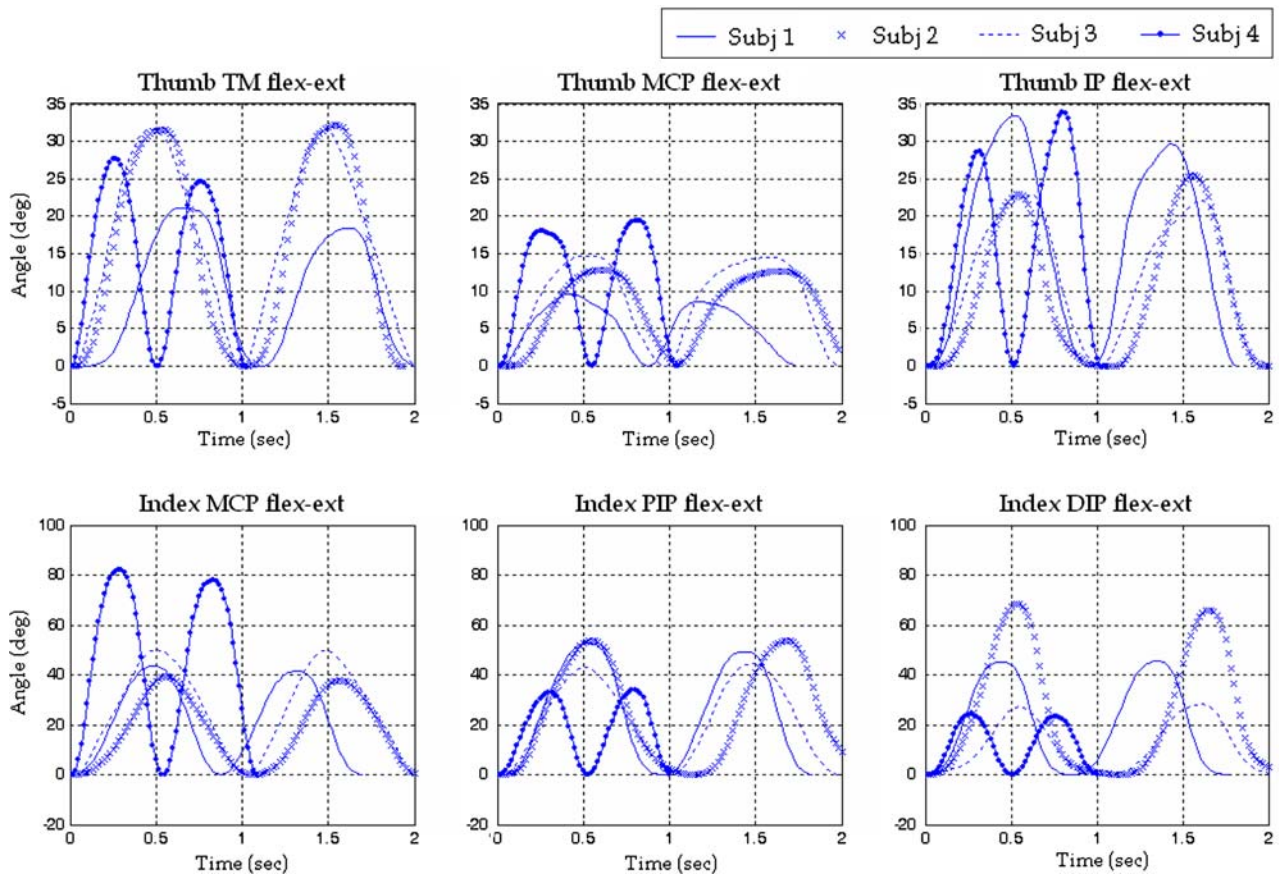


FIGURE 7. Flexion-extension angular data during no-timed tip-to-tip grasping. For all trajectories, only two tip-to-tip cycles are reported. Notice the inter-subject variability expressed in terms of execution speed and angular ranges of motion. However, all estimated angular trajectories are in the ranges of motion reported in literature.¹³

TABLE 4. Mean Bravais–Pearson coefficient of correlation among joint flexion angular profiles on a “tip to tip” grasping on 15 repetitions.

Movement	Subjects	Thumb			Index		
		TM	MCP	IP	MCP	PIP	DIP
“Tip to Tip” grasping 15-repetition	1	0.94	0.94	0.94	0.94	0.93	0.93
	2	0.96	0.94	0.94	0.96	0.96	0.98
	3	0.97	0.95	0.94	0.97	0.97	0.97
	4	0.97	0.94	0.96	0.97	0.97	0.95

TABLE 5. Percentage of residual occlusion during different motion sequence.

Movement	Acquisition frequency (Hz)	Frame number	Wrist (4 Mrk)		Palm (4 Mrk)		Thumb (4 Mrk)		Index (3 Mrk)	
			Occur.	Perc %	Occur.	Perc %	Occur.	Perc %	Occur.	Perc %
Fast typing	60	1270	16	0.31	16	0.31	16	0.31	435	11.42
	120	1154	0	0.00	0	0.00	2	0.04	0	0.00
Fast tip-to-tip grasping	60	778	0	0.00	0	0.00	125	4.47	137	5.87
	120	911	0	0.00	0	0.00	163	4.02	1	0.04
Slow thumb motion	60	942	0	0.00	0	0.00	30	0.80	0	0.00
	120	1295	0	0.00	1	0.02	0	0.00	0	0.00
Fast cylinder grasping	60	1398	0	0.00	2	0.04	87	1.56	106	2.53
	120	2098	0	0.00	2	0.02	0	0.00	0	0.00

TABLE 6. Computational load and real time performance.

Movement	Number of rendered frames	Time (Sec) = $t_{3D} + t_{lm} + t_r$	Mean display refresh frequency (Hz)
Fast typing	1270	27.753	45.76
Fast Tip-to-Tip Grasping	911	18.082	50.38
Slow thumb motion	1295	25.653	50.48
Fast cylinder grasping	2098	41.784	50.21

0.8%. Acquiring same movements at 120 Hz, a significant improvement of the results is obtained for task 1, task 3, and task 4 with all percentages approaching 0%. Task 2 exhibits improvement only for the index finger, whereas the tracking performance for the thumb markers does not sensibly improve (4.47 to 4.02%). This motor task seriously stressed both the tracking process, due to the high speed of execution (similar to a reflex) and the thumb kinematics, which is already demonstrated to be poorer than the one of the other fingers. Summarizing, inaccurate kinematics can affect markers tracking. On the other hand, it is not feasible to increase thresholds to recover lost markers, because of the high spatial resolution required to discriminate among close markers in tasks like grasping, where finger get close to each other.

Real-Time Performance

Results reported in (Table 6) show a reproduction frequency of the motion ranging from approximately 45 to 50 Hz confirming real-time capability of our system, where target considered was the real-time perception of the movement. The most critical time interval is the computational cost (approximately 20 ms) needed by the rendering process. The static labeling algorithm is computationally more time-consuming than the dynamic one: measured mean time values are approximately 1.0 and 0.7 ms, respectively (Fig. 8). However, both values are not critical and are sensibly lower than the computational load needed by the rendering process.

DISCUSSION AND CONCLUSIONS

As surface marker measurement technology progresses and reveals its value for non-invasive *in vivo* human hand motion studies,^{5,7,8,26,30,37} a methodology for determining the finger segmental kinematics has been missing. The present study represents an attempt to accomplish both the need for a plug-and-play marker scheme and for a real-time motion tracking. The success of deriving segmental kinematics based on a minimal set of surface markers, as demonstrated in the present study, lends support to standardization of

marker placement scheme for *in-vivo* investigating human hand motions. Marker placement design defined in this work, maximizing the marker separation and improving consistency, is a simplification of the one used in a previous study⁵ and resembles that proposed in.³⁷ In Ref. 5, the authors demonstrated that the estimation of AoRs and CoRs of the hand can be achieved with high accuracy using a controlled motion sequence with a repeatability in the range of 1° for AoRs estimation. Such results required a cumbersome marker protocol and off-line marker tracking. While implying a certain degree of inaccuracy (in the range of about 3.5° for the thumb and of 2.5° for other fingers), the simpler marker scheme, along with few anthropometrical measurements taken on the subject, allows the system to immediately reproduce real-time motion.

Similar analysis was performed for the other digits and errors are consistent with the index finger, so the authors chose to focus specifically on the index as representative of the other digits. So, without lack of generality, and supporting the hypothesis that all the markers on long fingers introduce the same level of error in calibrating the hand model, we focused the analysis on the thumb and the index fingers. Also, we analyzed tip-to-tip motion as representative and prototypical of hand structures. In particular, the calibration testing session showed high intra-subject repeatability² to different postures (RMS error was less than 0.5 mm) and adequate intra-subject repeatability to re-marking (RMS error was less than 2 mm) in agreement with literature.² Thumb re-marking was more critical than re-marking of the other fingers. Such a problem reflected on thumb kinematics results where, along with the lack of TM joint modeling, we reported lower accuracy. Reproducibility results on the estimated angular trajectories confirmed previous considerations.

In order to smooth out the effects of the inadequate thumb modeling, the intrinsic pronation direction of the metacarpus, through an additional marker, was included and estimated (Fig. 1b). The *in-vivo* estimated values of the pronation θ_{TM-PS} , which ranged approximately between 62° and 69°, proved to be

² Repeatability of the calibration was validated by analyzing the segmental lengths.

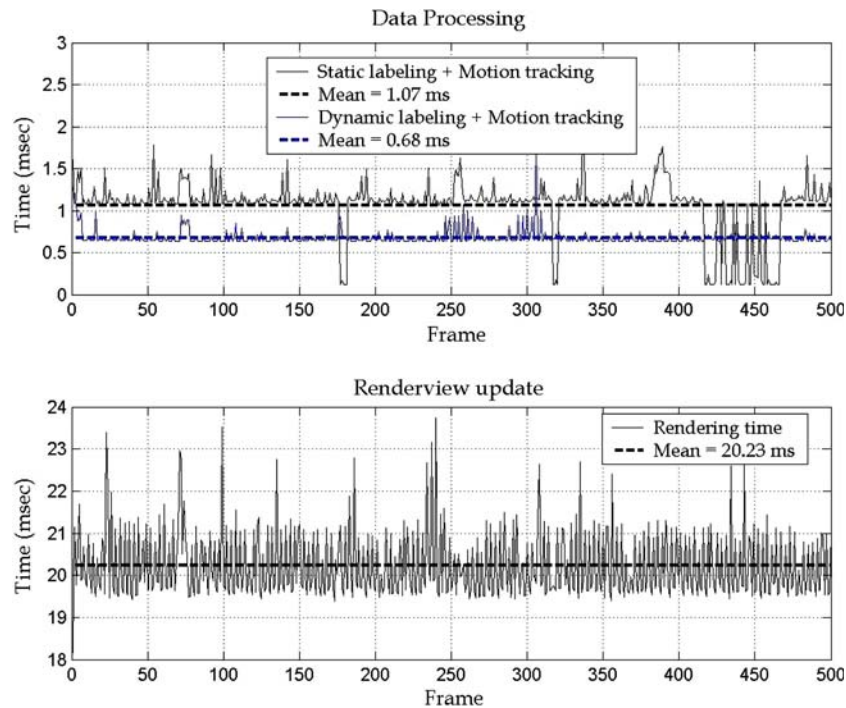


FIGURE 8. Real-time performances assessed in terms of computational load during grasping motion sequence (500 frames). In the first graph, data processing time (marker labeling plus motion tracking time) is computed while only static labeling is allowed (black) or allowing static-dynamic labeling interaction (blue). Note that peak values are uncorrelated and depend on the operating system activity. The minimum values in the black graph depend on the failure of the static labeling procedure: if the wrist markers are not tracked, no further labeling is attempted on any anatomical districts due to the hierarchy. The “Renderview update” graph represents the time needed by the graphic engine to render the virtual scene.

better with respect to the range values proposed in literature ($42^\circ < \theta_{TM-PS} < 48^\circ$). The use of the *in-vivo* estimated values reduced the error by a factor less than 35%.

Motion tracking procedure was tested at different acquisition frequencies (60 and 120 Hz) throughout different motor tasks. Most of typical articulated hand motions can be adequately reproduced at 60 Hz with at least six-cameras. In contrast, acquisition frequency of 120 Hz allowed the correct tracking of faster finger movements, as tip-to-tip and quick object grasping. The algorithm developed to recover marker occlusions proved extreme effectiveness to track fast hand motion where markers disappeared frequently. We finally demonstrated that the optimized code implementation guarantees real-time functionality in the range of 50 Hz.

In conclusion, the developed application allows calibration of 3D models customized on subject characteristics stored in the hand database, real-time model animation in the virtual scene, motion recording and batch visualization of movements for kinematic analysis, different markers configurations, extended report facilities about accuracy and repeatability results. As an example of immediate exploitation, the achieved performance offers the ability to use the application functionalities in the field of psychological and

neuropsychological experiments, where subjects have to perform hand motor task and visualize in real-time the virtual reproduction of its own motion. Future developments will concern the investigation of more sophisticated kinematic model description of the thumb and the integration of a dynamic calibration stage for a more accurate estimation of AoRs and CoRs. In addition, we are working towards the inclusion of true subject surface aspect in terms of morphology and visual appearance within the application, to be used during animation by range scanner and texture mapping.

ACKNOWLEDGMENT

The authors acknowledge the financial support from Italian Spatial Agency (ASI).

REFERENCES

- ¹An, K. N., E. Y. Chao, W. P. III Cooney, and R.L. Linscheid. Normative model of human hand for biomechanical analysis. *J. Biomech.* 12:775–788, 1979.

- ²Biggs, J., and K. Horch. A three-dimensional kinematic model of the human long finger and the muscles that actuate it. *Med. Eng. Phys.* 21:625–639, 1999.
- ³Braido, P., and X. Zhang. Quantitative analysis of finger motion coordination in hand manipulative and gestic acts. *Human Movement Sci.* 22:661–678, 2004.
- ⁴Buchholz, B., and T. J. Armstrong. A kinematic model of the human hand to evaluate its prehensile capabilities. *J. Biomech.* 25:149–162, 1992.
- ⁵Cerveri, P., N. Lopomo, A. Pedotti, and G. Ferrigno. Derivation of centers and axes of rotation for wrist and fingers in a hand kinematic model: robust methods and reliability results. *Ann. Biomed. Eng.* 33(3):401–411, 2005.
- ⁶Chèze, L., N. Doriot, M. Eckert, C. Rumelhart, and J. J. Comtet. Étude cinématique in vivo de l'articulation Trapèzométacarpienne. *Chirurgie de la Main* 20:23–30, 2001.
- ⁷Chiu, H. Y., F. C. Su, S. T. Wang, and H. Y. Hsu. The motion analysis system and goniometry of the finger joints. *J. Hand Surg.* 23B:788–791, 1998.
- ⁸Chiu, H. Y., S. C. Lin, F. C. Su, S. T. Wang, and H. Y. Hsu. The use of the motion analysis system for evaluation of loss of movement in the finger. *J. Hand Surg.* 25:195–199, 2000.
- ⁹Coert, J. H., H. G. van Dijke, S. E. Hovius, C. J. Snijders, and M. F. Meek. Quantifying thumb rotation during circumduction utilizing a video technique. *J. Orthopaedic Res.* 21:1151–1155, 2003.
- ¹⁰Crisco, J. J., X. Chen, M. M. Panjabi, and S. W. Wolfe. Optimal marker placement for calculating the instantaneous center of rotation. *J. Biomech.* 27:1183–1187, 1994.
- ¹¹Häger-Ross, C., and M. H. Schieber. Quantifying the independence of human finger movements: comparisons of digits, hands, and movement frequencies. *J. Neurosci.* 20(22):8542–8550, 2000.
- ¹²Hollister, A., W. L. Buford, L. M. Myers, D. J. Giurintano, and A. Novick. The axes of rotation of the thumb carpo-metacarpal joint. *J. Orthopaedic Res.* 10:454–460, 1992.
- ¹³Hollister, A., D. J. Giurintano, W. L. Buford, L. M. Myers, and A. Novick. The axes of rotation of the thumb interphalangeal and metacarpophalangeal joints. *Clin. Orthopaed.* 330:188–193, 1995.
- ¹⁴Leonard, L., D. Sirkett, G. Mullineux, G. E. B. Giddins, and A. W. Miles. Development of an in-vivo method of wrist joint motion analysis. *Clin. Biomech.* 20:166–171, 2005.
- ¹⁵Lin J., Y. Wu, and T. S. Huang. Modeling the constraints of human hand motion. *Human Motion, 2000. Proceedings. Workshop on. 7–8 Dec.*, 121–126, 2000.
- ¹⁶Kadaba, M. P., H. K. Ramakrishnan, M. E. Wootten, J. Gainey, G. Gorton, and C. V. B. Cochran. Repeatability of kinematic, kinetic, and electromyographic data in normal adult gait. *J. Orthopaed. Res.* 7:849–860, 1989.
- ¹⁷Kamper, D. G., E. G. Cruz, and M. P. Siegel. Stereotypical Fingertip Trajectories During Grasp. *J. Neurophys.* 90:3702–3710, 2003.
- ¹⁸Kuo, L.-C., W. P. Cooney III, M. Oyama, K. R. Kaufman, F.-C. Su, and K.-N. An. Feasibility of using surface markers for assessing motion of the thumb trapeziometacarpal joint. *Clin. Biomech.* 18:558–563, 2003.
- ¹⁹Kuo, L.-C., F.-C. Su, H.-Y. Chiu, and C.-Y. Yu. Feasibility of using a video-based motion analysis system for measuring thumb kinematics. *J. Biomech.* 35:1499–1506, 2002.
- ²⁰Maravita, A., C. Spence, and J. Driver. Multisensory integration and the body schema: close to hand and within reach. *Curr. Biol.* 13:531–539, 2003.
- ²¹Marin, F., H. Mannel, L. Claes, and L. Durselen. Accurate determination of a joint rotation center based on the minimal amplitude point method. *Comp. Aided Surg.* 8(1):30–34, 2003.
- ²²Mason, C. R., J. E. Gomez, and T. J. Ebner. Hand synergies during reach-to-grasp. *J. Neurophysiol.* 86(6):2896–2910, 2001.
- ²³Mason, C. R., L. S. Theverapperuma, C. M. Hendrix, and T. J. Ebner. Monkey hand postural synergies during reach-to-grasp in the absence of vision of the hand and object. *J. Neurophysiol.* 91(6):2826–2837, 2004.
- ²⁴Mason, C. R., C. M. Hendrix, and T. J. Ebner. Purkinje cells signal hand shape and grasp force during reach-to-grasp in the monkey. *J. Neurophysiol.* 95(1):144–158, 2006.
- ²⁵Novak, K. E., L. E. Miller, and J. C. Houk. Kinematic properties of rapid hand movements in a knob turning task. *Exp. Brain Res.* 132:419–433, 2000.
- ²⁶Rash, G. S., P. P. Belliappa, M. P. Wachowiak, N. N. Somia, and A. Gupta. A demonstration of the validity of a 3-D video motion analysis method for measuring finger flexion and extension. *J. Biomech.* 32:1337–1341, 1999.
- ²⁷Santos, V. J., and F. J. Valero-Cuevas. Reported anatomical variability naturally leads to multimodal distributions of Denavit–Hartenberg parameters for the human thumb. *IEEE Trans. Biomed. Eng.* 53(2):155–163, 2006.
- ²⁸Schieber, M. H., and M. Santello. Hand function: peripheral and central constraints on performance. *J. Appl. Phys.* 96:2293–2300, 2004.
- ²⁹Small, C. F., J. T. Bryant, and D. R. Pichora. Rationalization of kinematic descriptors for three-dimensional hand and finger motion. *J. Biomed. Eng.* 14(2):133–141, 1992.
- ³⁰Somia, N., G. S. Rash, M. Wachowiak, and A. Gupta. The initiation and sequence of digital joint motion. A three dimensional motion analysis. *J. Hand Surg.* 23B:792–795, 1998.
- ³¹Theverapperuma, L.S., C.M. Hendrix, C.R. Mason, and T.J. Ebner. Finger movements during reach-to-grasp in the monkey: amplitude scaling of a temporal synergy. *Exp. Brain Res.* 169(4):433–448, 2006.
- ³²Valero-Cuevas, F. J., M. E. Johanson, and J. D. Towles. Towards a realistic biomechanical model of the thumb: the choice of kinematics description may be more critical than the solution method or the variability/uncertainty of musculoskeletal parameters. *J. Biomech.* 36:1019–1030, 2003.
- ³³Weiss, A. P., D. C. Moore, C. Infantolino, J. J. Crisco, E. Akelman, and R. D. McGovern. Metacarpophalangeal joint mechanics after 3 different silicone arthroplasties. *J. Hand Surg.* 29(5):796–803, 2004.
- ³⁴Wu, G., F. C. T. van der Helm, H. E. J. Veeger, M. Makhsous, P. Van Roy, C. Anglin, J. Nagels, A. R. Karduna, K. McQuade, X. Wang, F. W. Werner, and B. Buchholz. ISB recommendation on definitions of joint coordinate systems of various joints for the reporting of human joint motion—Part II: shoulder, elbow, wrist and hand. *J. Biomech.* 32:1337–1341, 2005.
- ³⁵Yasumuro, Y., Q. Chen, and K. Chihara. Three-dimensional modelling of the human hand with motion constraints. *Image Vision Comput.* 17:149–156, 1999.
- ³⁶Yoshida, R., H. O. House, R. M. Patterson, M. A. Shah, and S. F. Viegas. Motion and morphology of the thumb metacarpophalangeal joint. *J. Hand Surg.* 28(5):753–757, 2003.
- ³⁷Zhang, X., S. W. Lee, and P. Braido. Determining finger segmental center of rotation in flexion-extension based on surface marker measurement. *J. Biomech.* 36:1097–1102, 2003.